

① Smart attendance checker

```
import java.util.Scanner;
```

```
public class SmartAttendanceChecker {  
    public static void main (String[] args) {  
        Scanner scanner = new Scanner (System.in);
```

```
        System.out.print ("Enter total number of classes  
        Conducted: ");
```

```
        int totalClass = scanner.nextInt();
```

```
        System.out.print ("Enter number of classes attended: ");  
        int attendedClass = scanner.nextInt();
```

```
        double attendancePercentage = ((double) attendedClasses  
        / totalClasses) * 100;
```

```
        System.out.printf ("Attendance Percentage: %.2f%%\n",  
        attendancePercentage);
```

```
        if (attendancePercentage >= 75 {
```

```
            System.out.println ("Status: Eligible for Exam");
```

```
        } else {
```

```
            System.out.println ("Status: Not Eligible");
```

```
        }
```

```
        scanner.close();
```

```
    }
```

```
}
```



```
Run SmartAttendanceChecker x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar"
Enter total number of classes conducted: 80
Enter number of classes attended: 65
Attendance Percentage: 81.25%
Status: Eligible for Exam
```

OOPtutorial.java > ooptutorial2.java > src > SmartAttendanceChecker

② AI age Predictor

```
import java.util.Scanner;
import java.util.LocalDate;

public class AIAgePredictor {
    public static void main (String[] args) {
        Scanner scanner = new Scanner (System.in);

        System.out.print ("Enter your current age: ");
        int currentAge = scanner.nextInt();

        int currentYear = LocalDate.now().getYear();

        System.out.println("\n--- Age Predictions ---");
        System.out.println("Your age after 10 years: " +
                           (currentAge + 10));
        System.out.println("Your age after 25 years: " +
                           (currentAge + 25));
        System.out.println("Your age after 50 years: " +
                           (currentAge + 50));

        // Bonus: Year they will turn 100
        int YearsTurn100 = 100 - currentAge;
        int year100 = currentYear + YearsTurn100;
        System.out.println("\nBonus Prediction:");
        System.out.println("You will turn 100 in the
        year: " + year100);
        scanner.close();
    }
}
```

Run AIAgePredictor x



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Enter your current age: 24

--- Age Predictions ---

Your age after 10 years : 34

Your age after 25 years : 49

Your age after 50 years : 74

OPtutorial.java > ooptutorial2.java > src > AIAgePredictor

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③ Fitness challenge tracker

```
import java.util.Scanner;
public class FitnessChallengeTracker {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int[] steps = new int[7];

        System.out.println("=== Fitness Challenge Tracker ===");
        System.out.println("Enter the number of steps walked  
for each of the 7 days: ");

        for (int i = 0; i < 7; i++) {
            System.out.print("Day " + (i+1) + ": ");
            steps[i] = scanner.nextInt();
        }

        int totalSteps = 0;
        int highestSteps = steps[0];

        for (int i = 0; i < 7; i++) {
            totalSteps += steps[i];
            if (steps[i] > highestSteps) {
                highestSteps = steps[i];
            }
        }

        double averageSteps = (double) totalSteps / 7;
        System.out.println("\n--- weekly summary ---");
        System.out.println("Total Steps: " + totalSteps);
        System.out.printf("Average Steps: %.2f\n", averageSteps);
        System.out.println("Highest steps: " + highestSteps);
        scanner.close();
    }
}
```

Run

FitnessChallengeTracker x



```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib"
=== Fitness Challenge Tracker ===
Enter the number of steps walked for each of the 7 days:
Day 1: [0.381s][error][attach] pipe write error (232)
500
Day 2: 550
Day 3: 600
Day 4: 700
Day 5: 750
Day 6: 800
Day 7: 850

--- Weekly Summary ---
Total Steps   : 4750
Average Steps : 678.57
Highest Steps : 850
```

This view is read-only

(v) Fibonacci series

```
import java.util.Scanner;
public static void main (String[] args) {
    Scanner scanner = new Scanner(System.in);
    System.out.print ("Enter the value of n (how many
        Fibonacci numbers to print): ");
    int n = scanner.nextInt();
    System.out.print ("Fibonacci series (first " + n
        + " numbers): ");
```

```
    int a = 0, b = 1;
```

```
    for (int i = 1; i <= n; i++) {
        System.out.print (a);
        if (i < n) System.out.print (", ");
```

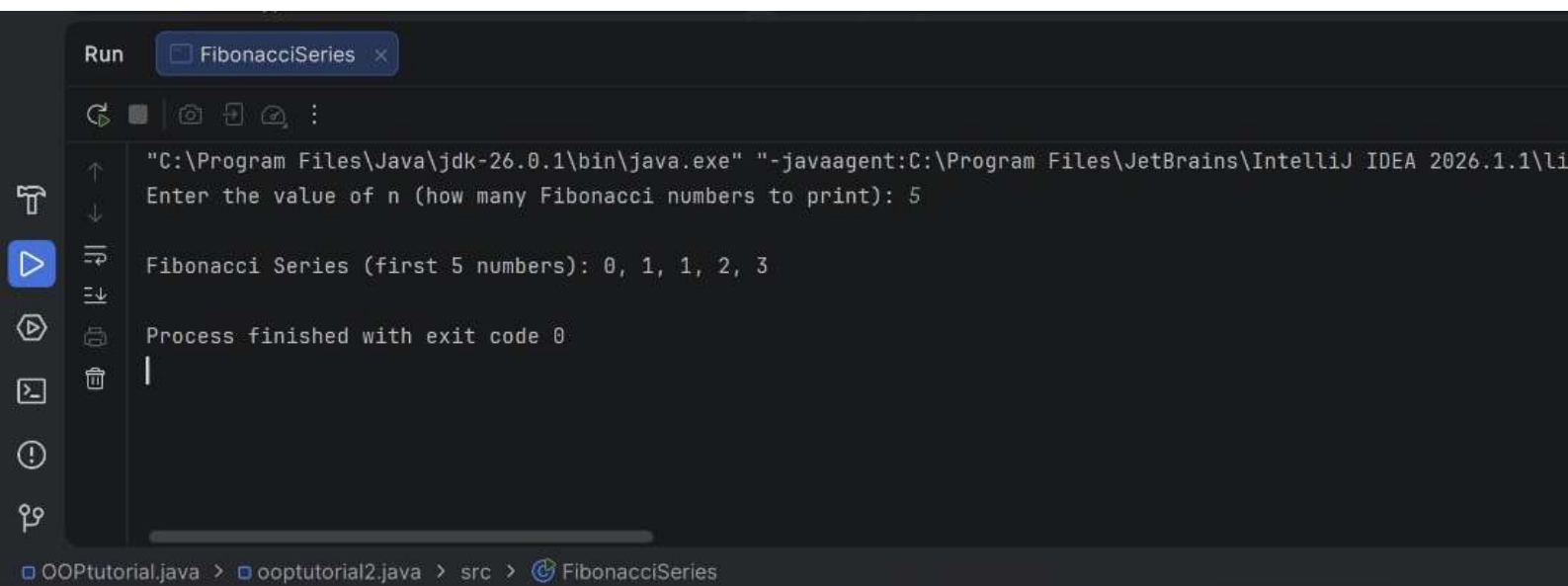
```
        int next = a + b;
        a = b;
        b = next;
```

```
    }
```

```
    System.out.println();
    scanner.close();
```

```
}
```

```
}
```



⑤ Prime number check

```
import java.util.Scanner
public class PrimeNumberCheck {
    public static boolean isPrime(int number) {
        if (number <= 1) return false;
        if (number <= 3) return true;
        if (number % 2 == 0 || number % 3 == 0) return false;
        for (int i = 5; i * i <= number; i += 6) {
            if (number % i == 0 || number % (i + 2) == 0) {
                return false;
            }
        }
        return true;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number to check if it is prime:");
        int number = scanner.nextInt();

        if (isPrime(number)) {
            System.out.println(number + " is a Prime Number.");
        } else {
            System.out.println(number + " is NOT a prime Number.");
        }
        scanner.close();
    }
}
```



The image shows the Run console window in IntelliJ IDEA. The window title is "Run PrimeNumberCheck x". The console output shows the execution of a Java program. The first line is the command: `"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib"`. The second line is the prompt: `Enter a number to check if it is prime: 67`. The third line is the output: `67 is a Prime Number.`. The fourth line is the status: `Process finished with exit code 0`. The console window has a toolbar on the left with icons for Run, Debug, and other actions.

```
Run PrimeNumberCheck x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib"
Enter a number to check if it is prime: 67
67 is a Prime Number.
Process finished with exit code 0
```


⑥ Logical operators check:

```
import java.util.Scanner;

public class LogicalOperatorCheck {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("=== Logical Operators check ===");
        System.out.print("Enter the first number : ");
        int a = scanner.nextInt();

        System.out.print("Enter the second number : ");
        int b = scanner.nextInt();

        System.out.print("Enter the third number : ");
        int c = scanner.nextInt();

        // Check if third number is the sum of first two using
        // logical operators
        if ((a + b == c) && (c - a == b) && (c - b == a)) {
            System.out.println("\nResult: YES! " + c + " is the
                sum of " + a + " and " + b + ".");
        } else {
            System.out.println("\nResult: NO! " + c + " is NOT
                the sum of " + a + " and " + b + ".");
            System.out.println("The actual sum of " + a + " and
                " + b + " is : " + (a + b));
        }

        scanner.close();
    }
}
```

Run

LogicalOperatorsCheck x

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"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib"

=== Logical Operators Check ===

Enter the first number : 20

Enter the second number : 10

Enter the third number : 30

Result: YES! 30 is the sum of 20 and 10.

Process finished with exit code 0

OOPTutorial.java > ooptutorial2.java > src > LogicalOperatorsCheck

7) Multiplication Table

```
import java.util.Scanner;
```

```
public class MultiplicationTable {  
    public static void main (String[] args) {  
        Scanner scanner = new Scanner (System.in);
```

```
        System.out.print ("Enter a number to print its  
        multiplication table: ");
```

```
        int number = scanner.nextInt();
```

```
        System.out.println ("\\n === Multiplication Table of  
        " + number + " ===");
```

```
        for (int i = 1; i <= 10; i++) {  
            System.out.printf ("%d x %d = %d \\n", num  
            i, number * i);  
        }
```

```
        scanner.close();
```

```
    }
```

```
}
```

Run

MultiplicationTable x

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"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib"

Enter a number to print its multiplication table: 11

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11 x 1 = 11
11 x 2 = 22
11 x 3 = 33
11 x 4 = 44
11 x 5 = 55
11 x 6 = 66
11 x 7 = 77
11 x 8 = 88
11 x 9 = 99
11 x 10 = 110

Process finished with exit code 0

|

⑧ Right-Angled Triangle pattern

```
import java.util.Scanner;  
public class RightAngledTriangle {  
    public static void main (String[] args) {  
        Scanner scanner = new Scanner(System.in);
```

```
        System.out.print("Enter the number of rows: ");  
        int rows = scanner.nextInt();
```

```
        System.out.println("\n === Right-Angled Triangle  
        Pattern === ");
```

```
        for (int i = 1; i <= rows; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print(" ");
```

```
            }
```

```
            scanner.close();
```

```
        }
```

```
    }
```

Run

RightAngledTriangle

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"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib"

Enter the number of rows: 7

=== Right-Angled Triangle Pattern ===

*

**

Process finished with exit code 0

Performance

Show Results

CPU

Heap Memory

OOPtutorial.java > ooptutorial2.java > src > RightAngledTriangle

classmate
Date _____
Page _____

2) Pyramid pattern

```
import java.util.Scanner;
```

```
public class PyramidPattern {  
    public static void main (String[] args) {  
        Scanner = new Scanner (System.in);
```

```
        System.out.print ("Enter the number of rows: ");  
        int rows = scanner.nextInt();
```

```
        System.out.println ("||n == Pyramid Pattern ==");
```

```
        for (int i = 1; i <= rows; i++) {
```

```
            for (int j = i; j <= rows; j++) {  
                System.out.print (" ");  
            }
```

```
            for (int k = 1; k <= (2 * i - 1); k++) {  
                System.out.print ("x");
```

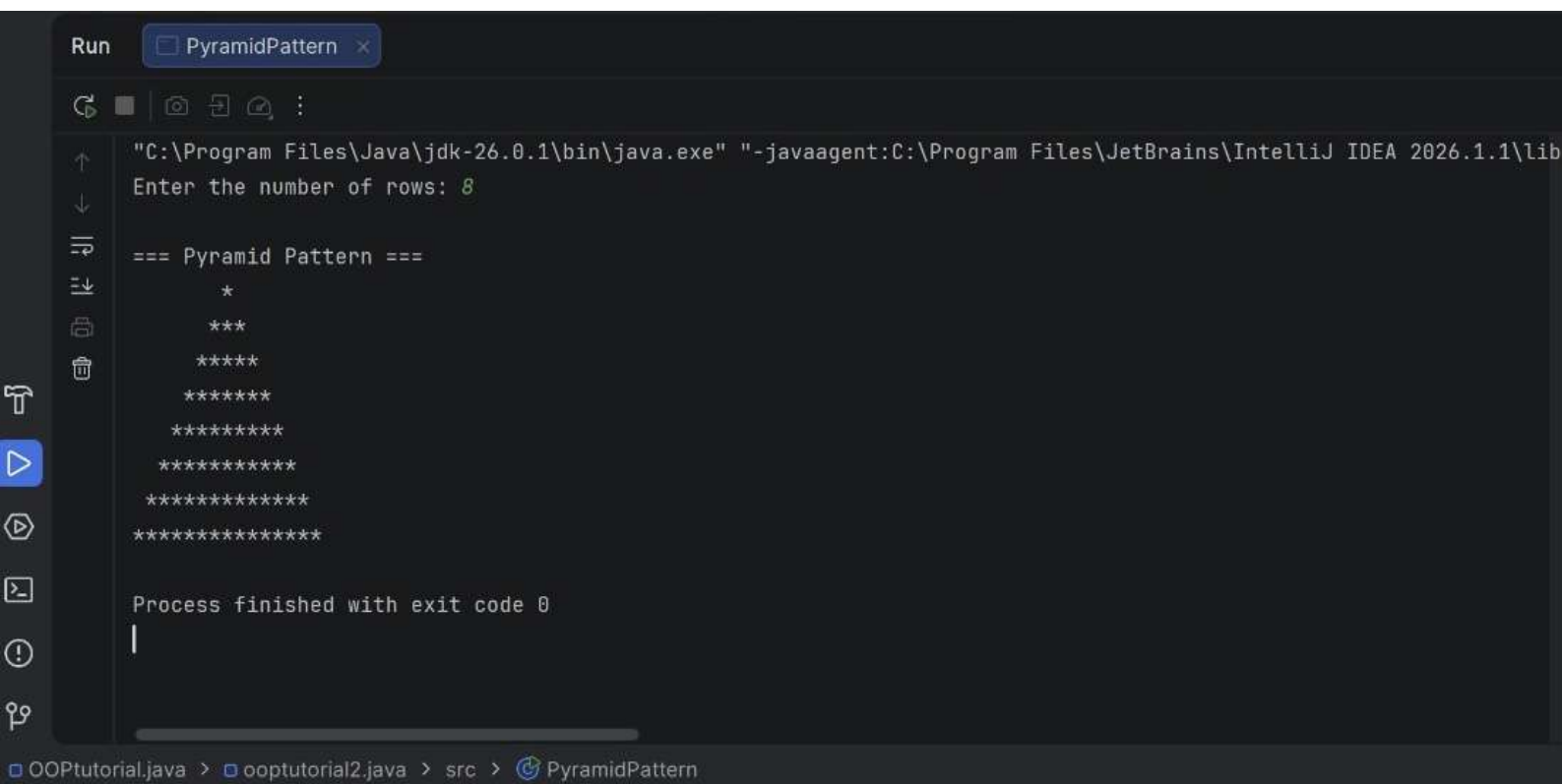
```
            }  
            System.out.println();
```

```
        }
```

```
        scanner.close();
```

```
    }
```

```
}
```



10) Palindrome Number

```
import java.util.Scanner;  
public class PalindromeNumber {  
    public static boolean isPalindrome(int x) {  
        if (x < 0) return false;  
        if (x < 10) return true;  
        if (x % 10 == 0) return false;
```

```
        int original = x;  
        int reversed = 0;  
        while (x > 0) {  
            int digit = x % 10;  
            reversed = reversed * 10 + digit;  
            x /= 10;
```

```
        }  
        return original == reversed;  
    }
```

```
    public static void main (String [] args) {  
        Scanner scanner = new Scanner (System.in);
```

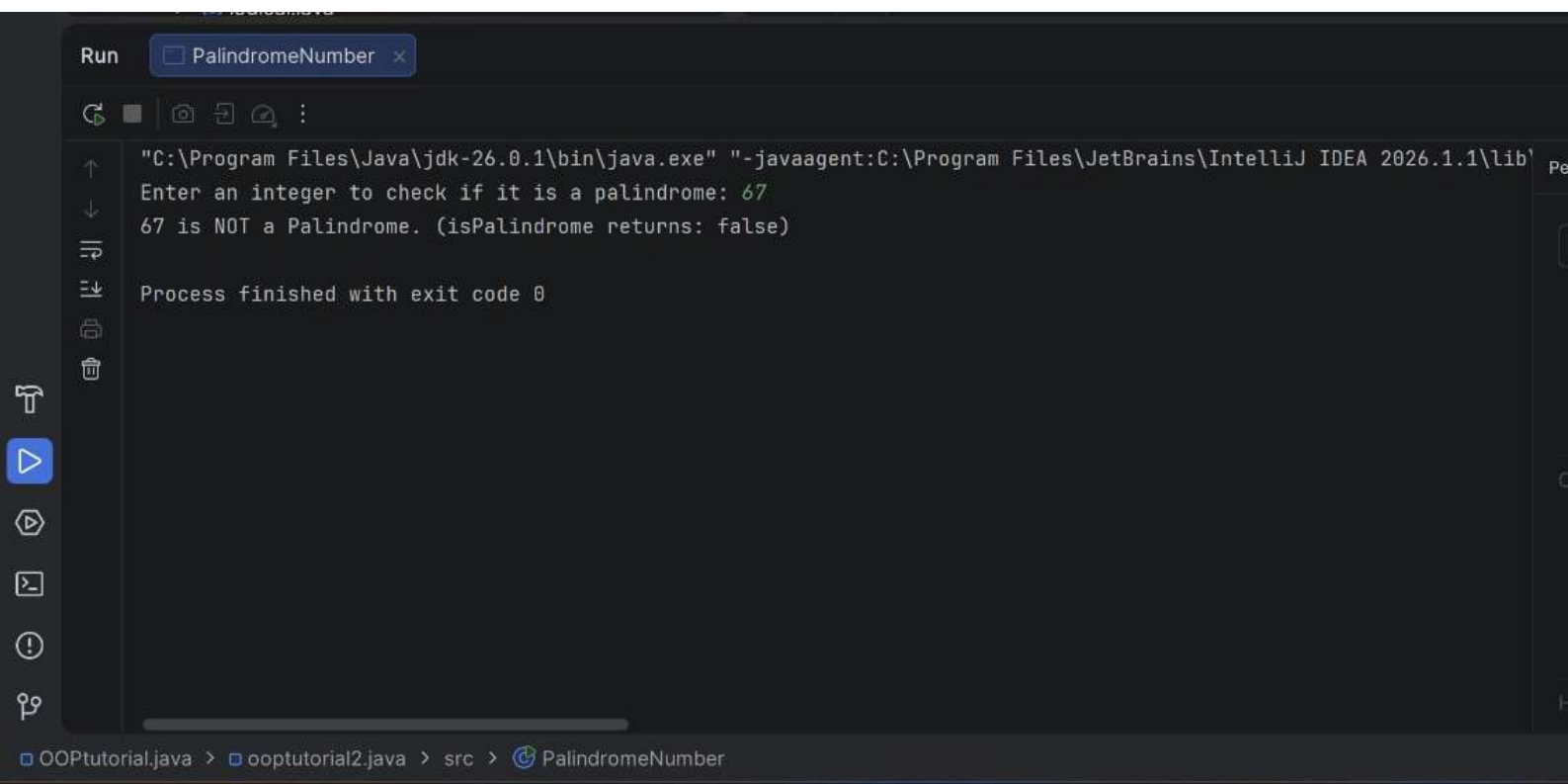
```
        System.out.print ("Enter an integer to check if  
        it is a palindrome? ");
```

```
        int number = scanner.nextInt();
```

```
        if (isPalindrome(number)) {  
            System.out.println (number + " is a palindrome.  
            (isPalindrome returns: true)");
```

```
        } else {  
            System.out.println (number + " is NOT a Palindrome.  
            (isPalindrome returns: false)");
```

```
        }  
        scanner.close();  
    }
```



(21)

Basic calculator.

```
import java.util.Scanner;
public static void main (String[] args) {
    Scanner scanner = new Scanner(System.in);
    System.out.println("=== Basic Calculator ===");
    System.out.print("Enter the first number:");
    double num1 = scanner.nextDouble();
    System.out.print("Enter the second number:");
    double num2 = scanner.nextDouble();
    System.out.println("\nSelect an operation:");
    System.out.println("1. Addition (+)");
    System.out.println("2. Subtraction (-)");
    System.out.println("3. Multiplication (*)");
    System.out.println("4. Division (/)");
    System.out.print("Enter your choice (1-4):");
    int choice = scanner.nextInt();
    double result;
    String operation;
    switch (choice) {
        case 1:
            result = num1 + num2;
            operation = "+";
            System.out.printf("\nResult: %.2f %s %.2f\n",
                num1, operation, num2, result);
            break;
        case 2:
            result = num1 - num2;
            operation = "-";
            System.out.printf("\nResult: %.2f %s %.2f\n",
                num1, operation, num2, result);
            break;
```


Case 3:

```
result = num1 * num2;
```

```
operation = "x";
```

```
System.out.printf ("In Result: %.2f %s %.2f =  
%.2f \n", num1, operation, num2, result);
```

```
break;
```

Case 4:

```
operation = "/";
```

```
if (num2 == 0) {
```

```
System.out.println ("In Error: Division by  
Zero is not allowed!");
```

```
} else {
```

```
result = num1 / num2;
```

```
System.out.printf ("In Result: %.2f %s %.2f =  
%.2f \n", num1, operation, num2, result);
```

```
}
```

```
break;
```

default:

```
System.out.println ("Invalid choice! Please  
select between 1 and 4.");
```

```
}
```

```
Scanner.close();
```

```
}  
}
```

```
}
```

Run BasicCalculator x

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"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib"

=== Basic Calculator ===

Enter the first number : 67

Enter the second number : 7

Select an operation:

1. Addition (+)

2. Subtraction (-)

3. Multiplication (*)

4. Division (/)

Enter your choice (1-4): +

(12) Ugrade calculator

```
import java.util.Scanner;
public class UgradeCalculator {
    Scanner scanner = new Scanner(System.in);
    System.out.println("=== Ugrade calculator (Lincoln University Ugrading System) ===");
    System.out.print("Enter the marks obtained (0-100): ");
    double marks = scanner.nextDouble();
    String grade = (marks >= 70) ? "First class (1st) - Distinction" :
        (marks >= 60) ? "Upper Second Class (2:1) - Merit" :
        (marks >= 50) ? "Lower Second Class (2:2) - Pass" :
        (marks >= 40) ? "Third class (3rd) - Pass" :
        "Fail";
    String remarks = (marks >= 70) ? "Exceptional performance!" :
        (marks >= 60) ? "very good performance." :
        (marks >= 50) ? "Satisfactory Performance." :
        (marks >= 40) ? "Minimum pass achieved." :
        "Below minimum pass mark. Retit required.";
    System.out.println("\n--- Ugrade Report ---");
    System.out.printf("Marks obtained : %.1f%%\n", marks);
    System.out.println("Classification : " + grade);
    System.out.println("Remarks : " + remarks);
    System.out.println("\n--- Lincoln University Ugrading scale ---");
    System.out.println("70% and above -> First Class (1st) / Distinction");
    System.out.println("60% - 69% -> Upper second class (2:1) / Merit");
    System.out.println("50% - 59% -> Lower second class (2:2) / Pass");
```



```
System.out.println("40% - 49% → Third class (3rd) /  
Pass");
```

```
System.out.println("Below 40% → Fail");
```

```
Scanner.close();
```

```
}
```

```
}
```


Run GradeCalculator x

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```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib'
=== Grade Calculator (Lincoln University Grading System) ===
Enter the marks obtained (0 - 100): 78

--- Grade Report ---
Marks Obtained : 78.0%
Classification : First Class (1st) - Distinction
Remarks      : Exceptional performance!

--- Lincoln University Grading Scale ---
70% and above -> First Class (1st) / Distinction
60% - 69%     -> Upper Second Class (2:1) / Merit
50% - 59%     -> Lower Second Class (2:2) / Pass
40% - 49%     -> Third Class (3rd) / Pass
Below 40%     -> Fail

Process finished with exit code 0
|
```

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